

Hemanthsai Katuri

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EDUCATION

Vignana Bharathi Institute of Technology

Bachelor of Technology in Electronics and Communication Engineering | CGPA: 7.2

Hyderabad, Telangana

2022 – 2026

PROFESSIONAL SUMMARY

Electronics and Communication Engineering undergraduate with experience in full-stack development, embedded systems, FPGA-based acceleration and scalable web platforms. Skilled in React.js, Next.js, Firebase, PostgreSQL and hardware-software co-design. Experienced in developing production-ready platforms, embedded architectures and research-oriented engineering solutions with strong problem-solving and performance optimization capabilities.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, C/C++, Verilog, HTML5, CSS3

Frontend: React.js, Next.js, Tailwind CSS, Framer Motion

Backend & Databases: Firebase, Supabase, PostgreSQL, PHP, MySQL, REST APIs, Node.js

Embedded: MCUs, MPUs, Arduino, ATmega328P, Communication Interfaces, FPGA Acceleration

UI/UX & Design: Figma, Responsive Design Principles, User Interface Design

Tools & Platforms: Git, GitHub, Vercel, VS Code, Arduino IDE, Google Apps Script, Xilinx Vitis HLS, GCC, Linux Server Administration, cPanel, phpMyAdmin

Concepts: Full Stack Development, Authentication, Technical SEO, Performance Optimization, Hardware-Software Co-Design

Operating Systems: Windows, Linux

EXPERIENCE

IEEE – VBIT Student Branch | Web Master

July 2025 – Present

Hyderabad, Telangana

- Re-engineered and deployed the official IEEE - VBIT Student Branch website using Next.js, TypeScript, and Tailwind CSS to improve scalability, responsiveness, and overall website performance.
- Engineered a secure PHP/MySQL admin dashboard for handling website feedback and query management with authentication and automated email routing.
- Improved Core Web Vitals and mobile responsiveness through optimized rendering strategies, lazy loading, and performance-focused frontend enhancements, achieving a Lighthouse performance score of 92.
- Implemented Technical SEO enhancements including structured metadata, sitemap optimization and dynamic OpenGraph integration for improved search visibility.

IEEE – VBIT Student Branch | Web Designer

July 2024 – July 2025

Hyderabad, Telangana

- Developed and maintained dynamic event registration platforms using JavaScript and Firebase, automating event management workflows for the student branch.
- Built secure admin dashboards enabling non-technical members to create and manage events, registrations and payment workflows in real-time.
- Integrated Razorpay payment gateway and automated registration data exports to Google Sheets using Google Apps Script.
- Designed and deployed interactive competition websites with multi-level progression systems and real-time validation using Cloud Firestore.

IEEE Robotics and Automation Society

March 2025 – April 2025

Virtual Internship — Research Methodology, Embedded Systems & IoT

- Completed a 5-week virtual internship focused on Research Methodology, 3D Modelling, Embedded Systems and IoT under IEEE Robotics and Automation Society (RAS) Hyderabad Section.

Hybrid Genetic Algorithm Framework for FPGA CNN Accelerators | *Xilinx Vitis HLS, CNN, GA*

- Developed a memory-aware optimization framework for FPGA-based CNN accelerators using Hybrid Genetic Algorithms and HLS directives.
- Implemented parameterized MAC-array and Conv2D accelerator models to analyze compute-memory trade-offs.
- Automated synthesis evaluation workflows for latency, DSP, LUT, BRAM and timing analysis in FPGA accelerator configurations.

Learnable Universal Remote Architecture | *ATmega328P, Arduino, EEPROM, IRremote*

- Designed and implemented a programmable universal IR remote supporting both standard and unknown IR protocols using ATmega328P.
- Implemented EEPROM-based persistent storage for learned IR commands with support for NEC and raw signal decoding.
- Developed a standalone hardware interface with runtime mode switching and dynamic IR learning capabilities.

Dynamic Event Registration Portal | *JavaScript, Firebase, Razorpay, Google Apps Script* | [Live](#)

- Developed a full-stack event registration system with real-time event management and payment integration for IEEE SB activities.
- Built secure Firebase-based admin panels supporting custom registration forms and automated attendee management.
- Integrated Razorpay payment gateway and automated Firestore-to-Google Sheets data synchronization workflows.

Svadhyay LMS | *Next.js, React, Tailwind CSS, Supabase, PostgreSQL* | [Live](#)

- Architected and developed a custom Learning Management System serving 390+ students with role-based dashboards for administrators and learners.
- Implemented secure authentication and database management using Supabase with PostgreSQL-backed access control.
- Engineered automated PDF certificate generation and scalable email notification systems for course workflows.

PUBLICATIONS & ACHIEVEMENTS

“A Low-Cost, Learnable Universal Remote Architecture with Persistent Storage for Unknown IR Protocols”

2025

- Published in IEEE Xplore. Proposed a cost-effective embedded system using ATmega328P with on-device learning and persistent storage for handling standard and unknown IR protocols.
- [IEEE Xplore Publication Link](#)

Achievements

- Secured **Global Rank 1848, Region 10 Rank 1048** and **University Rank 3** in **IEEEExtreme 18.0** as part of team **IEEEVBITSB9** in the international 24-hour competitive programming challenge organized by IEEE.
- Awarded **First Prize** in **AAKAR–2026** organized by the Research and Development Cell, Vignana Bharathi Institute of Technology, for technical project innovation and presentation excellence.